

SUPERSTORE SALES PERFORMANCE ANALYSIS USING SQL

SQL Portfolio Project

Prepared By:

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Tools & Technologies:

- MySQL
- SQL

Project Description:

This project analyses the Superstore retail dataset using SQL to evaluate sales performance, profitability, regional performance, customer segments, and discount impact. The analysis includes data cleaning, exploratory data analysis (EDA), and business-oriented SQL queries to generate actionable insights and recommendations.

Project Type:

Data Analyst Portfolio Project

Year: 2026

Executive Summary

This project presents a comprehensive sales performance analysis of the Superstore retail dataset using SQL. The objective of the analysis is to evaluate business performance across product categories, regions, customer segments, and discount strategies to identify opportunities for growth and profitability improvement.

The project begins with data validation and exploratory data analysis (EDA) to understand the dataset structure and key metrics. A series of business-oriented SQL queries were then executed to analyse sales trends, profit performance, regional contributions, customer segment behaviour, and the impact of discounts on profitability.

Key findings revealed significant variations in sales and profit across categories and regions, highlighting areas of strong performance as well as opportunities for improvement. The analysis also identified products and segments contributing the highest revenue and profit, along with factors affecting business profitability.

Based on the findings, strategic recommendations were proposed to support data-driven decision-making, improve operational efficiency, and enhance overall business performance. This project demonstrates the practical application of SQL for business analysis and decision support in a retail environment.

TABLE OF CONTENTS

Executive Summary	2
1. Introduction	4
2. Business Objectives	4
3. Dataset Overview	5
4. Data Cleaning & Validation	6
5. Exploratory Data Analysis (EDA)	8
6. Business Analysis Using SQL	10
6.1 Sales Performance Analysis	
6.2 Profitability Analysis	
6.3 Regional Analysis	
6.4 Customer Segment Analysis	
6.5 Discount Analysis	
6.6 Business Growth Insights	
7. Key Findings	30
8. Business Recommendations	33
9. Conclusion	35

1. Introduction

The retail industry generates large volumes of transactional data that can be analysed to understand business performance and support data-driven decision-making. This project focuses on analysing the Superstore dataset using SQL to evaluate sales, profitability, regional performance, customer segments, and discount strategies.

The dataset contains information related to product categories, sales revenue, profit, quantity sold, discounts, and geographic regions. By applying SQL queries and analytical techniques, meaningful insights can be extracted to identify trends, measure business performance, and uncover opportunities for improvement.

This project demonstrates the practical application of SQL in transforming raw business data into actionable insights that can support strategic decision-making in a retail environment.

2. Business Objectives

The primary objectives of this project are:

- Analyse overall sales and profitability performance.
- Identify top-performing and underperforming products and categories.
- Evaluate sales and profit contribution across different regions.
- Analyse customer segment performance and profitability.
- Assess the impact of discounts on business profit.
- Generate actionable business insights and recommendations based on data analysis.

3. Dataset Overview

The Superstore dataset is a retail sales dataset that contains transactional information related to product sales, profitability, discounts, customer segments, and regional performance. The dataset is widely used for business analysis and data analytics projects to identify trends, evaluate business performance, and support decision-making.

The dataset used in this project consists of the following attributes:

Column Name	Data Type	Description
Ship_Mode	Text	Shipping method used for delivery
Segment	Text	Customer segment classification
Country	Text	Country where the order was placed
City	Text	City where the order was placed
State	Text	State where the order was placed
Postal_Code	Numeric	Postal code of customer location
Region	Text	Sales region
Category	Text	Product category
Sub_Category	Text	Product sub-category
Sales	Numeric	Revenue generated from sales
Quantity	Numeric	Number of units sold
Discount	Numeric	Discount applied to the sale
Profit	Numeric	Profit earned from the sale

4. Data Cleaning & Validation

Before performing data analysis, the dataset was validated to ensure data quality, consistency, and accuracy. Data cleaning is an essential step in the analytics process as it helps identify missing values, duplicate records, and invalid data that may affect the reliability of the analysis.

The following validation checks were performed on the dataset:

- Verification of total records loaded into the database.
- Identification of missing values in critical columns.
- Detection of duplicate records.
- Validation of sales, quantity, and discount values.

Data Validation Checks Performed:

4.1 Total Records Verification

```
SELECT COUNT(*) AS Total_Records  
FROM ss_store;
```

Purpose: Verify that the dataset was imported successfully and contains the expected number of records.

4.2 Missing Values Check

```
SELECT *  
FROM ss_store  
WHERE Sales IS NULL  
OR Profit IS NULL  
OR Category IS NULL  
OR Region IS NULL;
```

Purpose: Ensure critical columns do not contain missing values.

4.3 Duplicate Records Check

```
SELECT *,  
        COUNT(*) AS Duplicate_Count  
FROM ss_store  
GROUP BY Ship_Mode, Segment, Country, City, State,  
        Postal_Code, Region, Category, Sub_Category,  
        Sales, Quantity, Discount, Profit  
HAVING COUNT(*) > 1;
```

Purpose: Identify duplicate records that may affect analysis results.

4.4 Invalid Data Check

```
SELECT *  
FROM ss_store  
WHERE Sales <= 0  
       OR Quantity <= 0  
       OR Discount < 0  
       OR Discount > 1;
```

Purpose: Verify that numerical fields contain valid business values.

Data Cleaning Summary:

The dataset was successfully validated through record verification, missing value checks, duplicate record identification, and data quality assessment. No major data quality issues were identified that could significantly impact the analysis. Therefore, the dataset was considered suitable for further exploratory data analysis and business intelligence reporting.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the overall structure and characteristics of the dataset. The objective of this step was to summarize key metrics, identify data patterns, and gain an initial understanding of the business before conducting detailed analysis.

5.1 Total Records

SQL Query:

```
SELECT COUNT(*) AS Total_Records  
FROM ss_store;
```

Output: 9,994 records

Insight: The dataset contains '9994' records, providing sufficient data for meaningful sales and profitability analysis.

5.2 Total Sales

SQL Query:

```
SELECT SUM(Sales) AS Total_Sales  
FROM ss_store;
```

Output: \$2,297,200.86 Total Sales

Insight: The business generated total sales of '\$2,297,200.86', indicating the overall revenue earned during the analysis period.

5.3 Total Profit

```
SELECT SUM(Profit) AS Total_Profit  
FROM ss_store;
```

Output: \$286,397.02 Total Profit

Insight: Total profit earned was '\$286,397.02'.

5.4 Total Quantity Sold

```
SELECT SUM(Quantity) AS Total_Quantity  
FROM ss_store;
```

Output: \$37,873.00 Total Quantity

Insight: A total of '\$37,873.00' units were sold.

5.5 Product & Geographic Coverage

```
SELECT COUNT(DISTINCT Category) AS Categories,  
       COUNT(DISTINCT Sub_Category) AS Sub_Categories,  
       COUNT(DISTINCT State) AS States,  
       COUNT(DISTINCT Region) AS Regions  
FROM ss_store;
```

Output: Categories: 3 | Sub_Categories: 17 | States: 49 | Regions: 4

Insight: The dataset covers multiple product categories and geographic regions

EDA Summary:

The exploratory analysis provided an overview of the dataset, including total sales, profit, quantity sold, and business coverage across categories and regions. These findings established a foundation for the detailed business analysis performed in the next section.

6. Business Analysis Using SQL

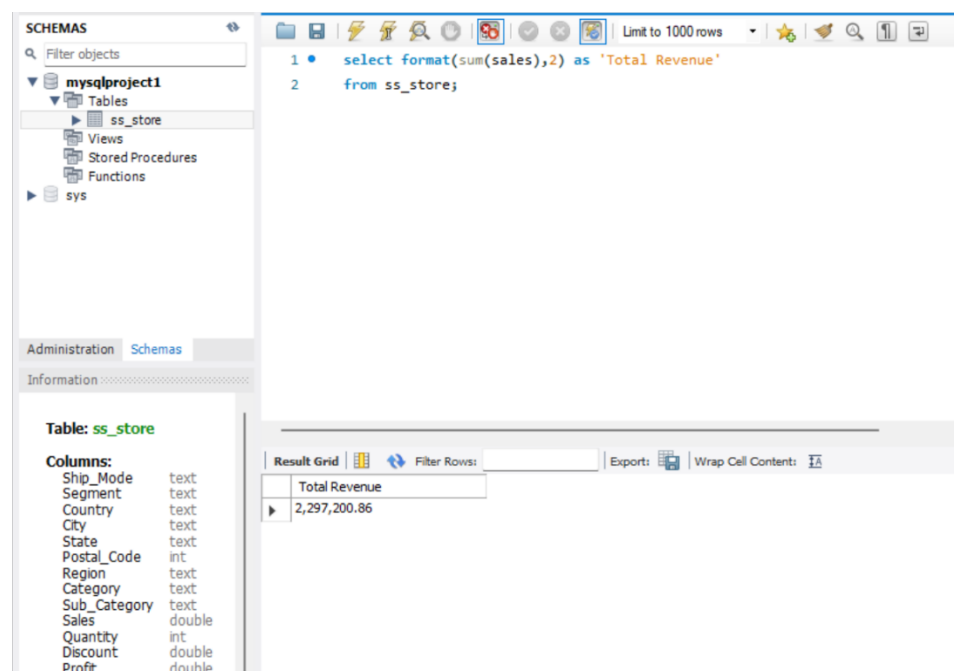
This section focuses on analysing business performance using SQL queries. The analysis covers sales performance, profitability, regional trends, customer segment behaviour, discount impact, and business growth insights. The objective is to transform raw data into actionable business insights that support data-driven decision-making.

6.1 Sales Performance Analysis

Sales performance analysis focuses on evaluating the revenue generated by the business across different products and categories. Understanding sales trends helps identify top-performing products, key revenue contributors, and areas that require improvement. This analysis provides valuable insights into business growth and overall market performance.

Query 1: What is the overall revenue generated by the business?

Output:



The screenshot displays a MySQL IDE interface. On the left, the 'SCHEMAS' panel shows a tree view with 'mysqlproject1' expanded, containing 'Tables', 'Views', 'Stored Procedures', and 'Functions'. The 'ss_store' table is selected under 'Tables'. Below this, the 'Table: ss_store' section lists columns and their data types: Ship_Mode (text), Segment (text), Country (text), City (text), State (text), Postal_Code (int), Region (text), Category (text), Sub_Category (text), Sales (double), Quantity (int), Discount (double), and Profit (double). The main query editor shows a SQL query:

```
1 select format(sum(sales),2) as 'Total Revenue'
2 from ss_store;
```

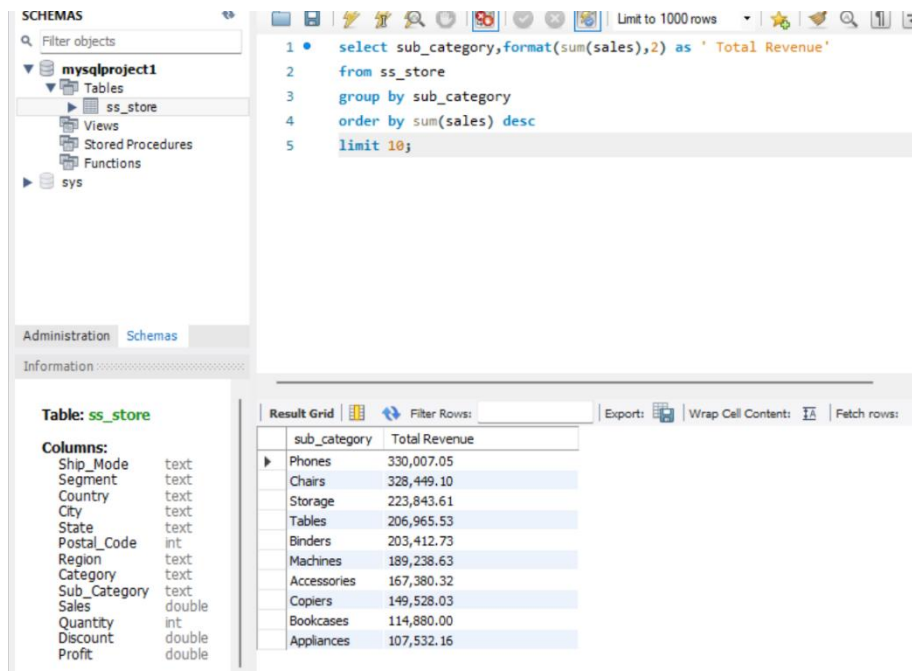
 The 'Result Grid' at the bottom shows a single row with the column 'Total Revenue' and the value '2,297,200.86'.

Total Revenue
2,297,200.86

Business Insight: The business generated a total revenue of '2,297,200.86', indicating the overall sales performance across all products and regions.

Query 2: Which are the top 10 highest revenue-generating products?

Output:



The screenshot shows a SQL IDE interface. On the left, a 'SCHEMAS' panel displays a tree view with 'mysqlproject1' expanded, showing 'Tables', 'Views', 'Stored Procedures', 'Functions', and 'sys'. The 'ss_store' table is selected. Below this, the 'Table: ss_store' information is shown, listing columns and their data types: Ship_Mode (text), Segment (text), Country (text), City (text), State (text), Postal_Code (int), Region (text), Category (text), Sub_Category (text), Sales (double), Quantity (int), Discount (double), and Profit (double). The main area displays a SQL query:

```
1 select sub_category,format(sum(sales),2) as 'Total Revenue'
2 from ss_store
3 group by sub_category
4 order by sum(sales) desc
5 limit 10;
```

 Below the query, the 'Result Grid' shows the top 10 sub-categories by total revenue. The columns are 'sub_category' and 'Total Revenue'.

sub_category	Total Revenue
Phones	330,007.05
Chairs	328,449.10
Storage	223,843.61
Tables	206,965.53
Binders	203,412.73
Machines	189,238.63
Accessories	167,380.32
Copiers	149,528.03
Bookcases	114,880.00
Appliances	107,532.16

Business Insight: The identified products contribute significantly to overall revenue and represent key drivers of business growth.

Query 3: Which products contribute the least to total sales?

Output:

Table: ss_store

Columns:

Ship_Mode	text
Segment	text
Country	text
City	text
State	text
Postal_Code	int
Region	text
Category	text
Sub_Category	text
Sales	double
Quantity	int
Discount	double
Profit	double

Result Grid:

sub_category	Total Revenue
Fasteners	3,024.28
Labels	12,486.31
Envelopes	16,476.40
Art	27,118.79
Supplies	46,673.54

Business Insight: These products generate the lowest sales and may require performance evaluation or marketing improvements.

Query 4: What percentage of total revenue comes from each category?

Output:

Table: ss_store

Columns:

Ship_Mode	text
Segment	text
Country	text
City	text
State	text
Postal_Code	int
Region	text
Category	text
Sub_Category	text
Sales	double
Quantity	int
Discount	double
Profit	double

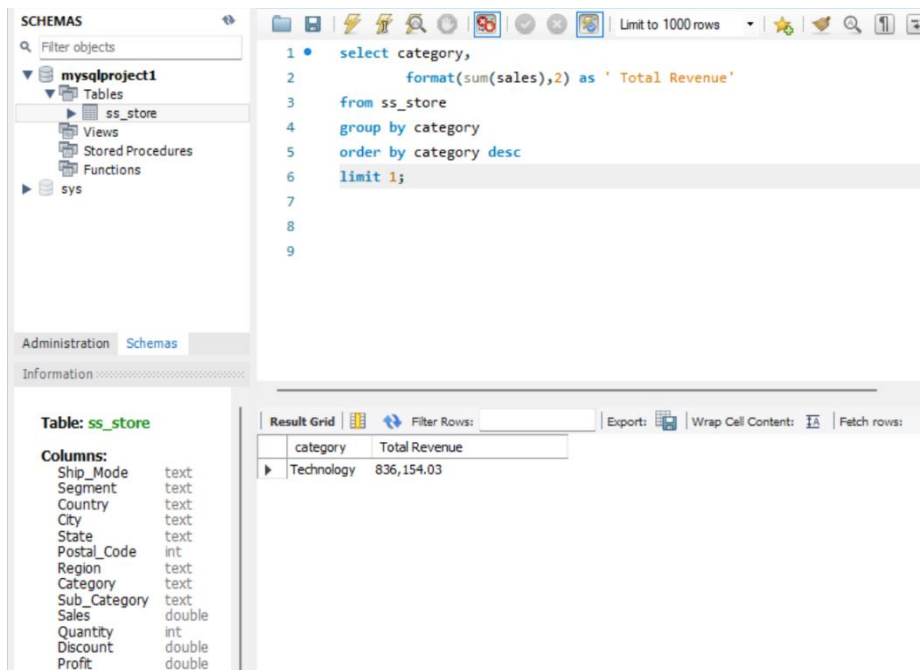
Result Grid:

category	Total Revenue
Furniture	741,999.80
Office Supplies	719,047.03
Technology	836,154.03

Business Insight: This analysis highlights the revenue contribution of each product category.

Query 5: Which category contributes the most revenue to the business?

Output:



The screenshot displays a SQL IDE interface. On the left, a 'SCHEMAS' pane shows a database named 'mysqlproject1' with a table 'ss_store'. The main editor shows a SQL query:

```
1 select category,
2     format(sum(sales),2) as 'Total Revenue'
3 from ss_store
4 group by category
5 order by category desc
6 limit 1;
```

 Below the editor, a 'Table: ss_store' section lists columns: Ship_Mode (text), Segment (text), Country (text), City (text), State (text), Postal_Code (int), Region (text), Category (text), Sub_Category (text), Sales (double), Quantity (int), Discount (double), and Profit (double). To the right, a 'Result Grid' shows the query output:

category	Total Revenue
Technology	836,154.03

Business Insight: The highest-performing category serves as a major contributor to overall business revenue.

Sales Analysis Summary:

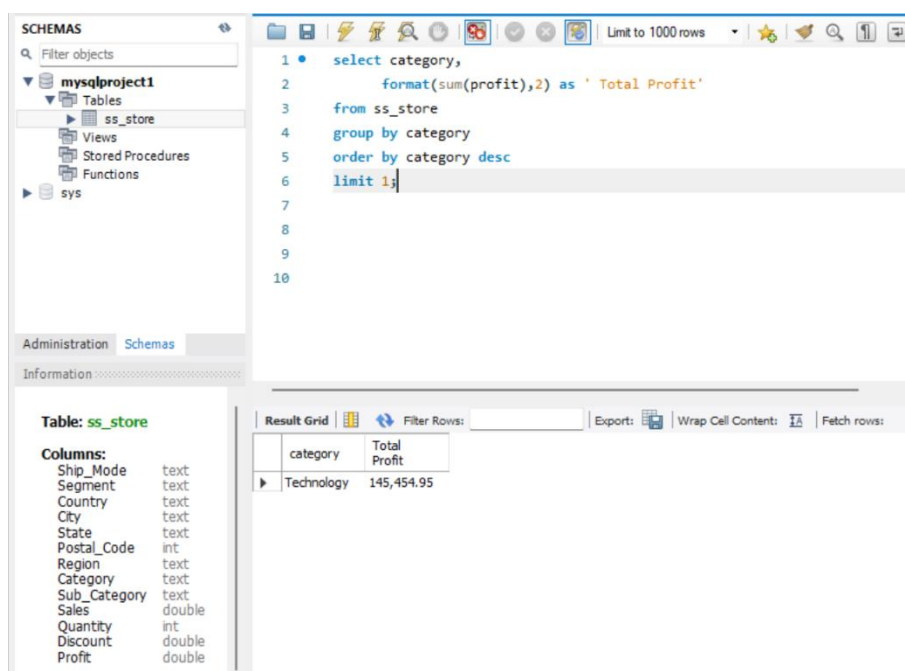
The sales analysis identified the primary revenue-generating products and categories, providing insights into the key drivers of business performance.

6.2 Profitability Analysis

Profitability analysis helps evaluate how effectively the business converts sales into profit. This section identifies profitable categories, loss-making products, and key profit contributors to support better business decisions.

Query 6: Which category generates the highest profit?

Output:



The screenshot shows a MySQL IDE interface. On the left, the 'SCHEMAS' panel displays a tree view with 'mysqlproject1' expanded, showing 'Tables', 'Views', 'Stored Procedures', and 'Functions'. The 'ss_store' table is selected. Below this, the 'Table: ss_store' information is shown, listing columns: Ship_Mode (text), Segment (text), Country (text), City (text), State (text), Postal_Code (int), Region (text), Category (text), Sub_Category (text), Sales (double), Quantity (int), Discount (double), and Profit (double). The main editor displays a SQL query:

```
1 • select category,  
2     format(sum(profit),2) as 'Total Profit'  
3   from ss_store  
4  group by category  
5  order by category desc  
6  limit 1;
```

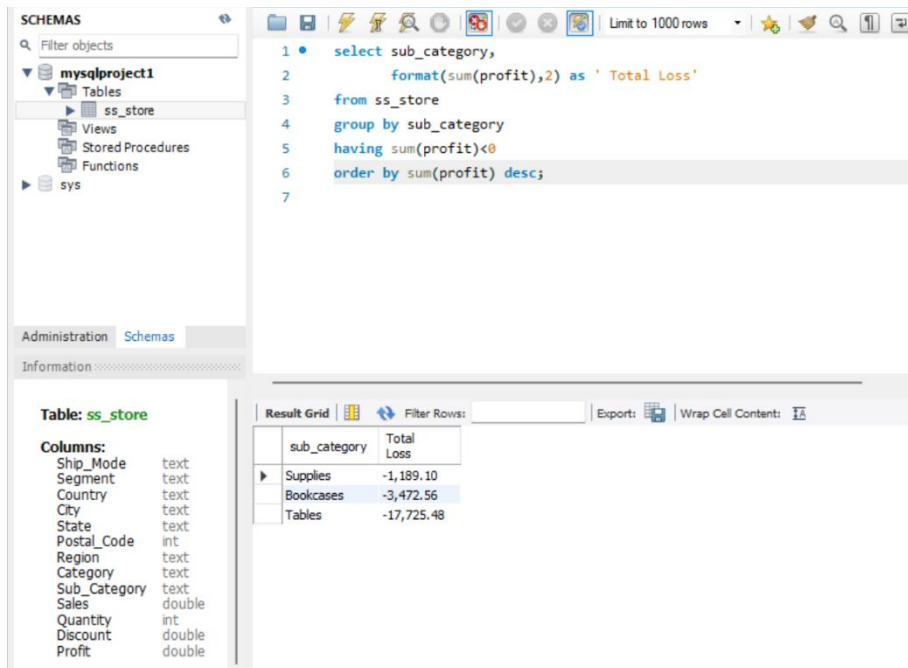
Below the query editor, the 'Result Grid' shows the output of the query:

category	Total Profit
Technology	145,454.95

Business Insight: The category generating the highest profit represents the most financially successful product group within the business.

Query 7: Which sub-categories are operating at a loss?

Output:



```
1 • select sub_category,
2       format(sum(profit),2) as 'Total Loss'
3   from ss_store
4  group by sub_category
5  having sum(profit)<0
6  order by sum(profit) desc;
7
```

Table: **ss_store**

Columns:

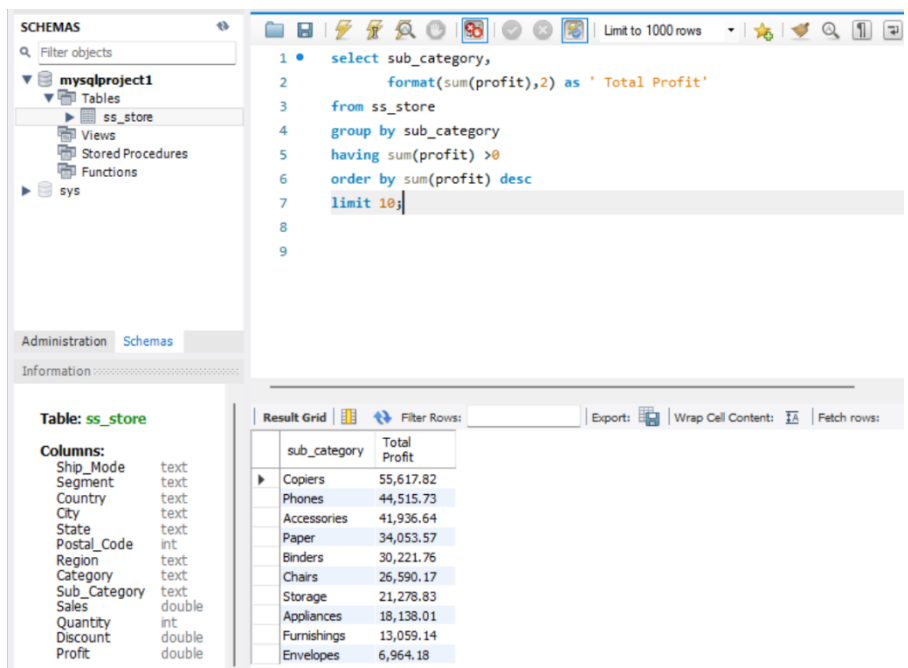
Ship_Mode	text
Segment	text
Country	text
City	text
State	text
Postal_Code	int
Region	text
Category	text
Sub_Category	text
Sales	double
Quantity	int
Discount	double
Profit	double

sub_category	Total Loss
Supplies	-1,189.10
Bookcases	-3,472.56
Tables	-17,725.48

Business Insight: These sub-categories generate negative profit and may require pricing, discount, or operational review.

Query 8: What are the top 10 most profitable products?

Output:



```
1 • select sub_category,
2       format(sum(profit),2) as 'Total Profit'
3   from ss_store
4  group by sub_category
5  having sum(profit) >0
6  order by sum(profit) desc
7  limit 10;
8
9
```

Table: **ss_store**

Columns:

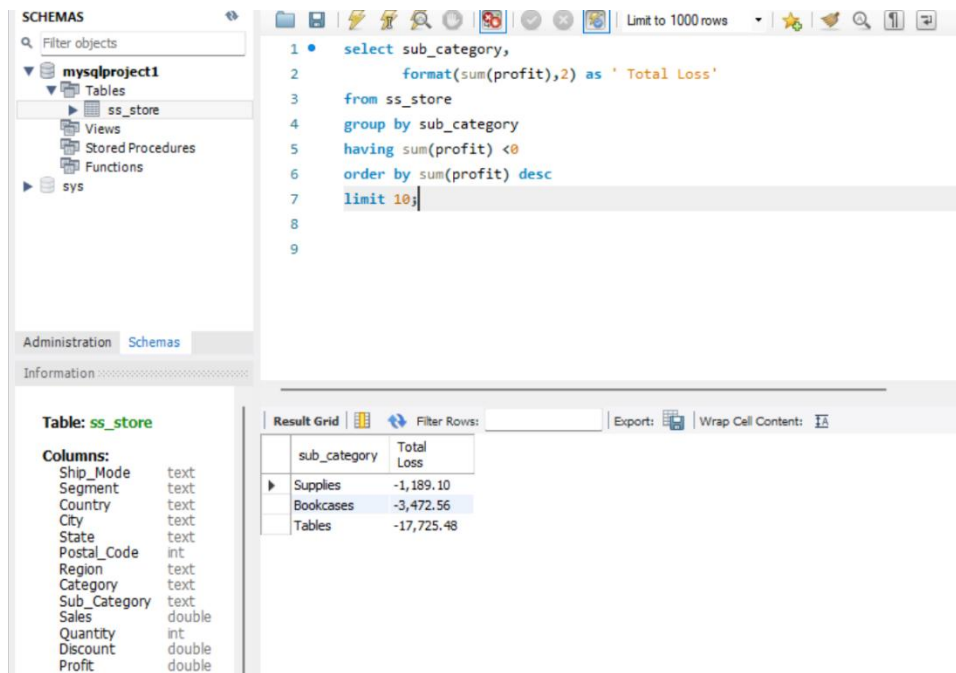
Ship_Mode	text
Segment	text
Country	text
City	text
State	text
Postal_Code	int
Region	text
Category	text
Sub_Category	text
Sales	double
Quantity	int
Discount	double
Profit	double

sub_category	Total Profit
Copiers	55,617.82
Phones	44,515.73
Accessories	41,936.64
Paper	34,053.57
Binders	30,221.76
Chairs	26,590.17
Storage	21,278.83
Appliances	18,138.01
Furnishings	13,059.14
Envelopes	6,964.18

Business Insight: These products contribute significantly to overall profitability and should be prioritized in sales and marketing strategies.

Query 9: What are the top 10 loss-making products?

Output:



The screenshot displays a MySQL IDE interface. On the left, the 'SCHEMAS' panel shows the database structure, including 'mysqlproject1', 'Tables', 'Views', 'Stored Procedures', 'Functions', and 'sys'. The 'ss_store' table is selected. The main editor shows a SQL query:

```
1 • select sub_category,  
2       format(sum(profit),2) as 'Total Loss'  
3 from ss_store  
4 group by sub_category  
5 having sum(profit) <0  
6 order by sum(profit) desc  
7 limit 10;  
8  
9
```

Below the query editor, the 'Result Grid' shows the results of the query:

sub_category	Total Loss
Supplies	-1,189.10
Bookcases	-3,472.56
Tables	-17,725.48

The 'Information' panel on the left lists the columns and data types for the 'ss_store' table:

Table: ss_store

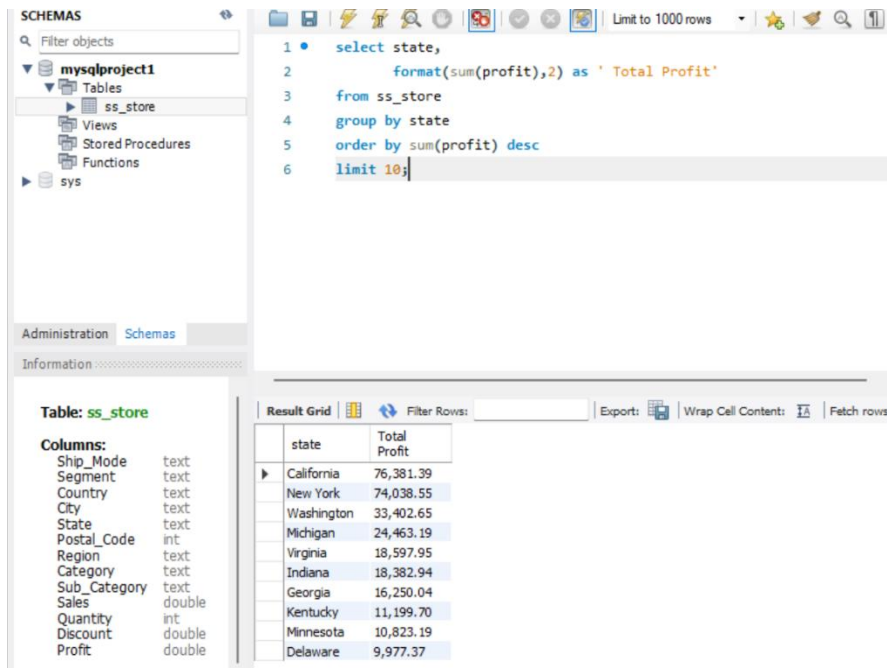
Columns:

- Ship_Mode: text
- Segment: text
- Country: text
- City: text
- State: text
- Postal_Code: int
- Region: text
- Category: text
- Sub_Category: text
- Sales: double
- Quantity: int
- Discount: double
- Profit: double

Business Insight: These products negatively impact business profitability and may require corrective actions.

Query 10: Which states contribute the highest profit?

Output:



The screenshot shows a SQL IDE interface. On the left, a 'SCHEMAS' pane displays a tree view with 'mysqlproject1' expanded, showing 'Tables', 'Views', 'Stored Procedures', and 'Functions'. The 'ss_store' table is selected. Below this, the 'Table: ss_store' structure is shown with columns: Ship_Mode (text), Segment (text), Country (text), City (text), State (text), Postal_Code (int), Region (text), Category (text), Sub_Category (text), Sales (double), Quantity (int), Discount (double), and Profit (double). The main editor displays a SQL query:

```
1 select state,
2     format(sum(profit),2) as 'Total Profit'
3 from ss_store
4 group by state
5 order by sum(profit) desc
6 limit 10;
```

 The bottom pane shows the 'Result Grid' with 10 rows of data. The columns are 'state' and 'Total Profit'. The data is sorted by total profit in descending order.

state	Total Profit
California	76,381.39
New York	74,038.55
Washington	33,402.65
Michigan	24,463.19
Virginia	18,597.95
Indiana	18,382.94
Georgia	16,250.04
Kentucky	11,199.70
Minnesota	10,823.19
Delaware	9,977.37

Business Insight: The identified states represent the strongest contributors to business profit and can be targeted for future growth opportunities.

Profitability Analysis Summary:

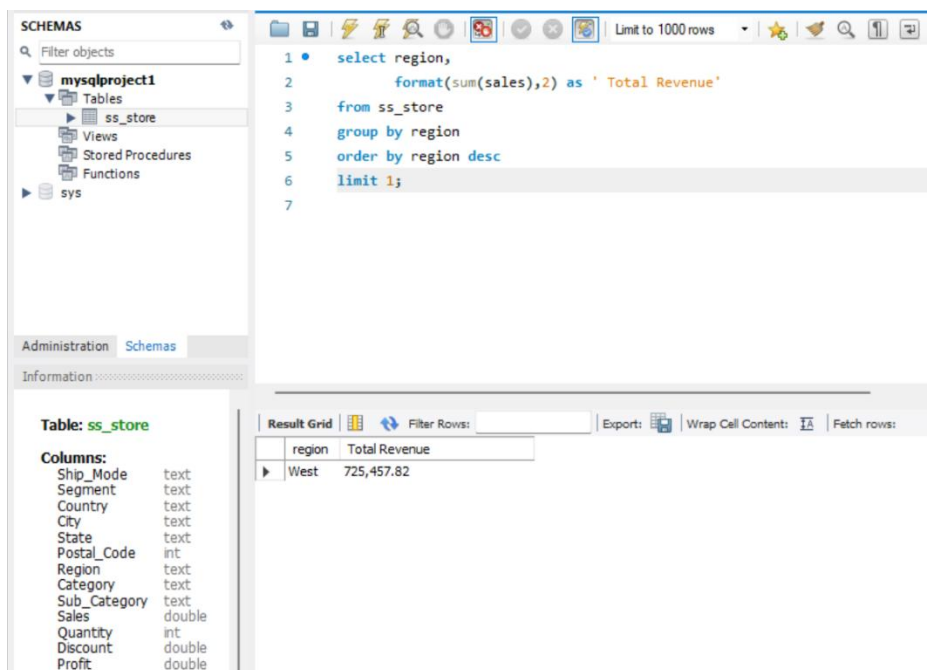
The profitability analysis identified the most profitable categories, products, and states while highlighting areas generating losses. These insights help management focus on high-performing products and improve profitability in underperforming areas.

6.3 Regional Analysis

Regional analysis helps evaluate business performance across different geographical locations. Understanding regional sales and profit trends enables organizations to identify high-performing markets and regions that require strategic improvement.

Query 11: Which region generates the highest sales revenue?

Output:



The screenshot displays a SQL IDE interface. On the left, the 'SCHEMAS' panel shows a tree view with 'mysqlproject1' expanded, containing 'Tables', 'Views', 'Stored Procedures', 'Functions', and 'sys'. The 'ss_store' table is selected under 'Tables'. Below this, the 'Table: ss_store' structure is listed with columns and their data types: Ship_Mode (text), Segment (text), Country (text), City (text), State (text), Postal_Code (int), Region (text), Category (text), Sub_Category (text), Sales (double), Quantity (int), Discount (double), and Profit (double). The main editor shows a SQL query:

```
1 select region,
2     format(sum(sales),2) as 'Total Revenue'
3 from ss_store
4 group by region
5 order by region desc
6 limit 1;
```

 The 'Result Grid' at the bottom shows the output of the query:

region	Total Revenue
West	725,457.82

Business Insight: The region with the highest sales revenue represents the strongest market in terms of customer demand and revenue generation.

Query 12: Which region generates the highest profit?

Output:

Table: ss_store

Columns:

Ship_Mode	text
Segment	text
Country	text
City	text
State	text
Postal_Code	int
Region	text
Category	text
Sub_Category	text
Sales	double
Quantity	int
Discount	double
Profit	double

```
1 select region,
2     format(sum(profit),2) as 'Total Profit'
3 from ss_store
4 group by region
5 order by region
6 limit 1;
```

region	Total Profit
Central	39,706.36

Business Insight: The most profitable region contributes significantly to overall business profitability and demonstrates effective market performance.

Query 13: Which states are consistently generating losses?

Output:

Table: ss_store

Columns:

Ship_Mode	text
Segment	text
Country	text
City	text
State	text
Postal_Code	int
Region	text
Category	text
Sub_Category	text
Sales	double
Quantity	int
Discount	double
Profit	double

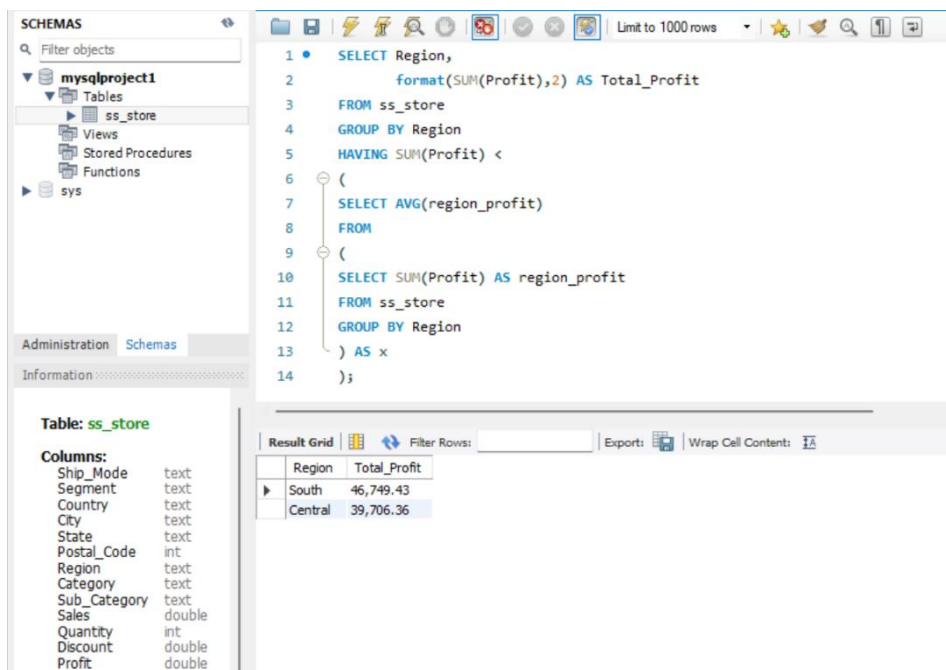
```
1 select state,
2     format(sum(profit),2) as 'Total Loss'
3 from ss_store
4 group by state
5 having sum(profit) < 0
6 order by sum(profit) desc;
```

state	Total Loss
Oregon	-1,190.47
Florida	-3,399.30
Arizona	-3,427.92
Tennessee	-5,341.69
Colorado	-6,527.86
North Carolina	-7,490.91
Illinois	-12,607.89
Pennsylvania	-15,559.96
Ohio	-16,971.38
Texas	-25,729.36

Business Insight: These states generate negative profit and may require detailed investigation into pricing, discounts, operational costs, or market conditions.

Query 14: Which regions have profit lower than the company average?

Output:



The screenshot displays a SQL IDE interface. On the left, the 'SCHEMAS' pane shows a tree view with 'mysqlproject1' expanded, containing 'Tables', 'Views', 'Stored Procedures', 'Functions', and 'sys'. The 'ss_store' table is selected under 'Tables'. Below this, the 'Table: ss_store' structure is shown with columns: Ship_Mode (text), Segment (text), Country (text), City (text), State (text), Postal_Code (int), Region (text), Category (text), Sub_Category (text), Sales (double), Quantity (int), Discount (double), and Profit (double). The main editor shows a SQL query:

```
1 SELECT Region,
2     format(SUM(Profit),2) AS Total_Profit
3 FROM ss_store
4 GROUP BY Region
5 HAVING SUM(Profit) <
6 (
7     SELECT AVG(region_profit)
8     FROM
9     (
10        SELECT SUM(Profit) AS region_profit
11        FROM ss_store
12        GROUP BY Region
13    ) AS x
14 );
```

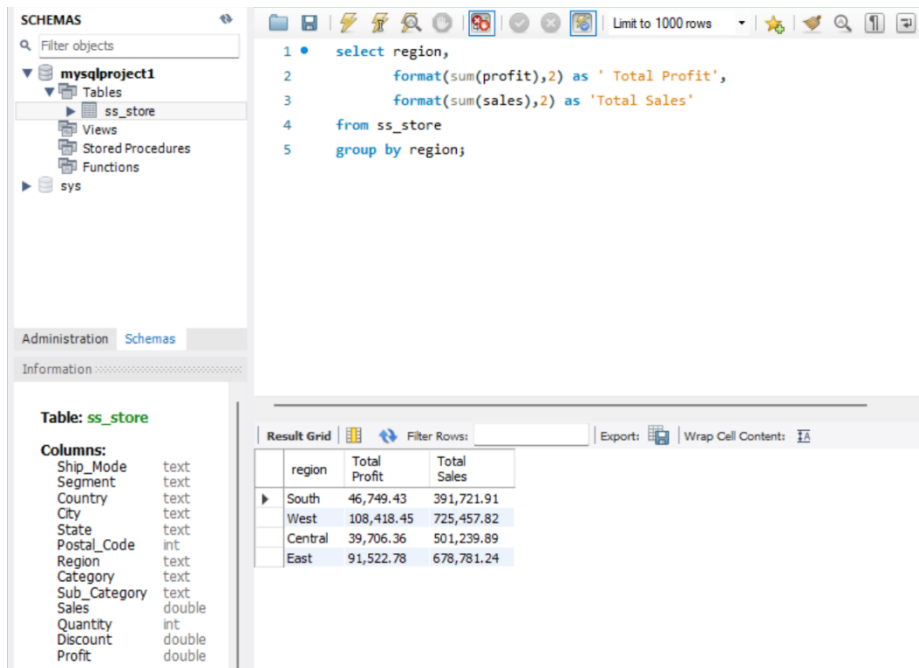
At the bottom, the 'Result Grid' shows the output of the query:

Region	Total_Profit
South	46,749.43
Central	39,706.36

Business Insight: Regions performing below the company average indicate opportunities for business improvement and targeted strategic actions.

Query 15: What is the sales and profit contribution of each region?

Output:



The screenshot shows a MySQL IDE interface. On the left, the 'SCHEMAS' panel displays a tree view with 'mysqlproject1' expanded, showing 'Tables' and 'ss_store'. Below this, the 'Table: ss_store' is detailed with its columns and data types. The main editor shows a SQL query:

```
1 select region,
2     format(sum(profit),2) as 'Total Profit',
3     format(sum(sales),2) as 'Total Sales'
4 from ss_store
5 group by region;
```

 The 'Result Grid' at the bottom displays the query results in a table format.

region	Total Profit	Total Sales
South	46,749.43	391,721.91
West	108,418.45	725,457.82
Central	39,706.36	501,239.89
East	91,522.78	678,781.24

Business Insight: Comparing sales and profit across regions helps identify markets that generate strong revenue while maintaining healthy profitability.

Regional Analysis Summary:

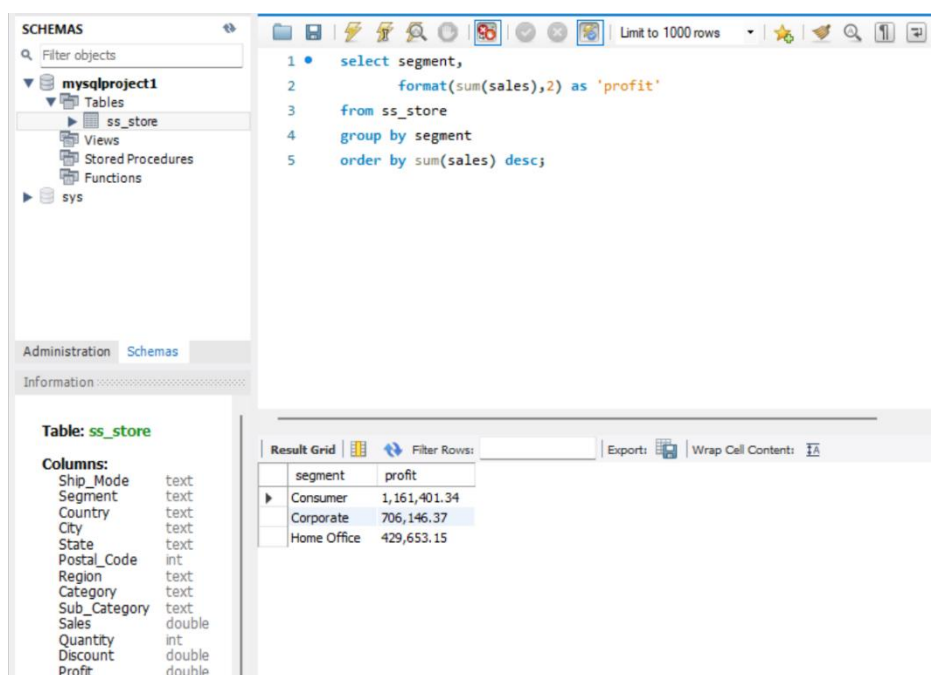
The regional analysis highlighted variations in sales and profit performance across different geographic locations. The findings identified top-performing regions as well as areas requiring attention, providing valuable insights for regional business planning and resource allocation.

6.4 Customer Segment Analysis

Customer segment analysis helps identify which customer groups contribute the most to sales and profitability. Understanding customer behaviour enables businesses to allocate resources effectively, improve customer satisfaction, and maximize long-term value.

Query 16: Which customer segment generates the highest revenue?

Output



The screenshot shows a SQL IDE interface. On the left, the 'SCHEMAS' pane displays a tree view for 'mysqlproject1' with folders for 'Tables', 'Views', 'Stored Procedures', 'Functions', and 'sys'. The 'Tables' folder is expanded, showing 'ss_store'. Below this, the 'Table: ss_store' is detailed with its columns and data types: Ship_Mode (text), Segment (text), Country (text), City (text), State (text), Postal_Code (int), Region (text), Category (text), Sub_Category (text), Sales (double), Quantity (int), Discount (double), and Profit (double). The main editor displays a SQL query:

```
1 select segment,
2     format(sum(sales),2) as 'profit'
3 from ss_store
4 group by segment
5 order by sum(sales) desc;
```

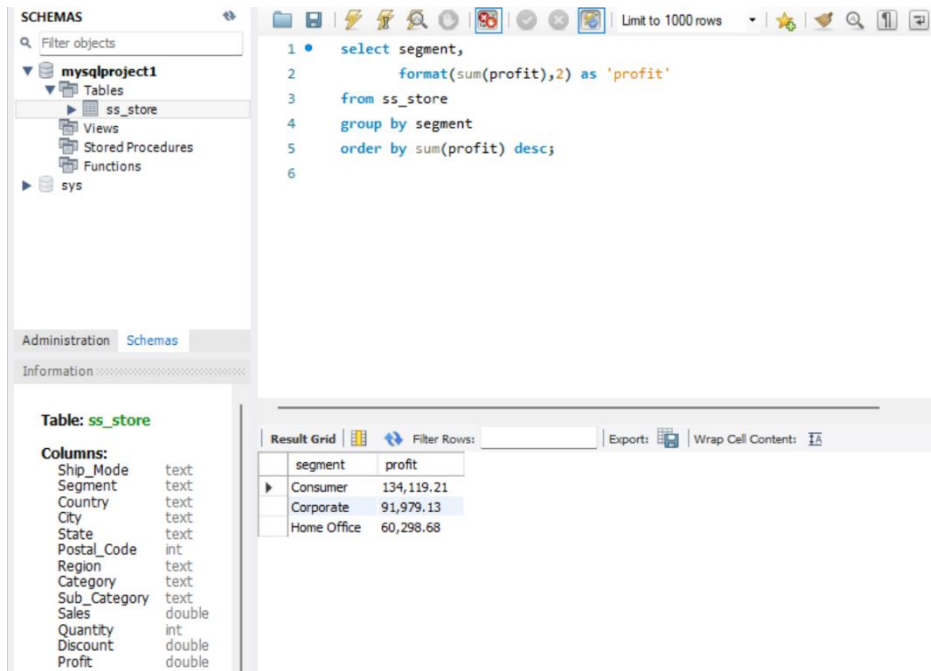
Below the query editor, the 'Result Grid' shows the output of the query:

segment	profit
Consumer	1,161,401.34
Corporate	706,146.37
Home Office	429,653.15

Business Insight: The highest revenue-generating segment represents the largest contributor to overall business sales and customer demand.

Query 17: Which customer segment generates the highest profit?

Output:



The screenshot shows a database query tool interface. On the left, the 'SCHEMAS' pane displays the 'mysqlproject1' database with a table 'ss_store'. The 'Columns' list for 'ss_store' includes: Ship_Mode (text), Segment (text), Country (text), City (text), State (text), Postal_Code (int), Region (text), Category (text), Sub_Category (text), Sales (double), Quantity (int), Discount (double), and Profit (double). The main query editor shows the following SQL:

```
1 select segment,
2     format(sum(profit),2) as 'profit'
3 from ss_store
4 group by segment
5 order by sum(profit) desc;
```

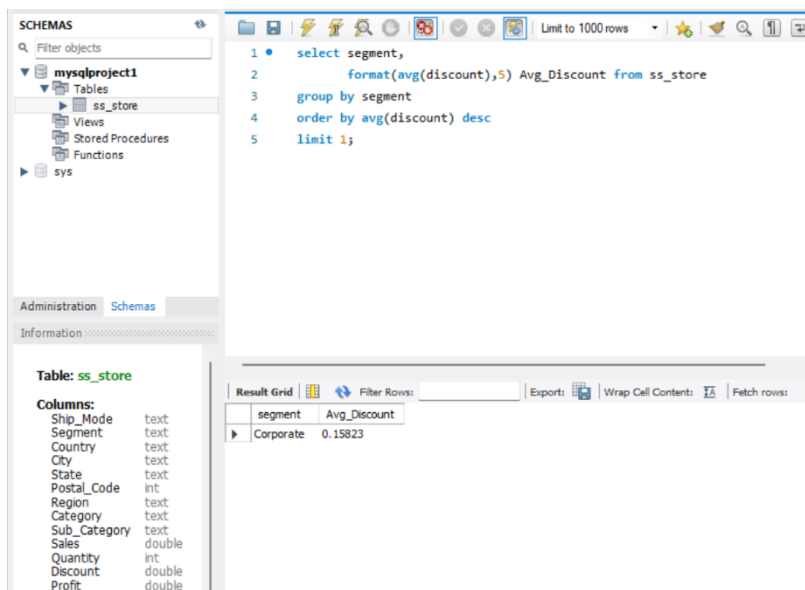
The 'Result Grid' shows the following data:

segment	profit
Consumer	134,119.21
Corporate	91,979.13
Home Office	60,298.68

Business Insight: This segment contributes the most to business profitability and represents a valuable customer group.

Query 18: Which segment receives the highest average discount?

Output:



The screenshot shows the same database query tool interface. The 'Columns' list for 'ss_store' is the same as in the previous screenshot. The main query editor shows the following SQL:

```
1 select segment,
2     format(avg(discount),5) Avg_Discount from ss_store
3 group by segment
4 order by avg(discount) desc
5 limit 1;
```

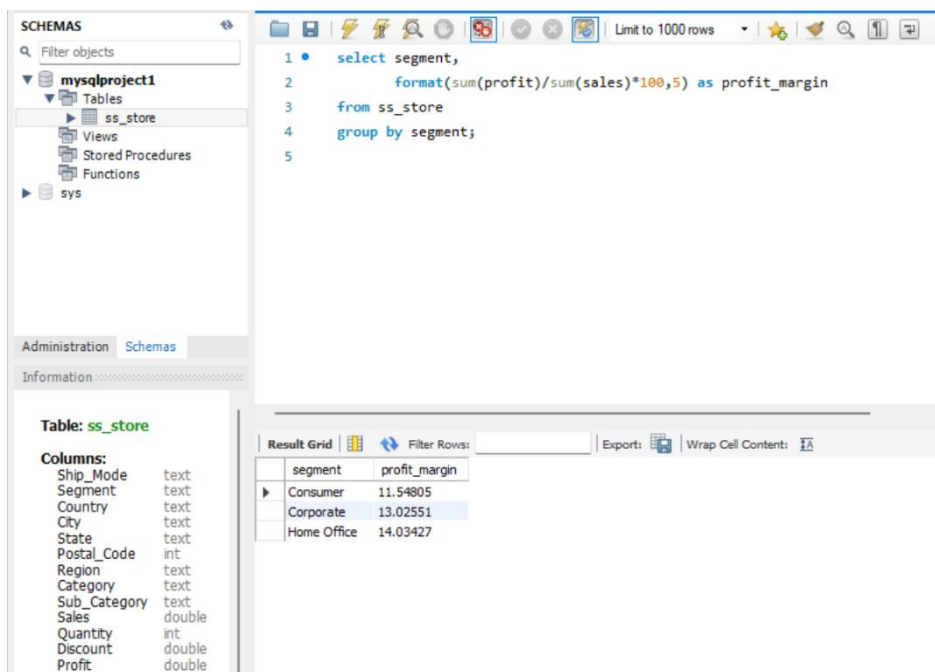
The 'Result Grid' shows the following data:

segment	Avg_Discount
Corporate	0.15823

Business Insight: The segment receiving the highest average discount may require evaluation to ensure discounts are not negatively affecting profitability.

Query 19: What is the profit margin for each customer segment?

Output:



The screenshot shows a MySQL IDE interface. On the left, the 'SCHEMAS' panel displays a tree view for 'mysqlproject1' with folders for 'Tables', 'Views', 'Stored Procedures', 'Functions', and 'sys'. The 'ss_store' table is selected under 'Tables'. Below this, the 'Table: ss_store' structure is shown with columns and their data types: Ship_Mode (text), Segment (text), Country (text), City (text), State (text), Postal_Code (int), Region (text), Category (text), Sub_Category (text), Sales (double), Quantity (int), Discount (double), and Profit (double). The main editor displays a SQL query:

```
1 select segment,
2     format(sum(profit)/sum(sales)*100,5) as profit_margin
3 from ss_store
4 group by segment;
5
```

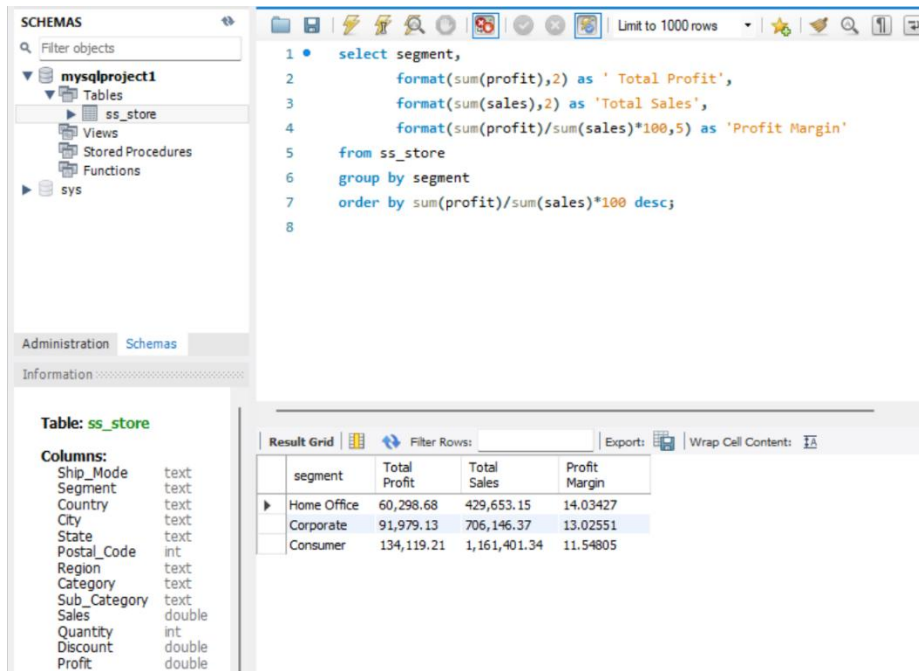
 The 'Result Grid' at the bottom shows the output of the query:

segment	profit_margin
Consumer	11.54805
Corporate	13.02551
Home Office	14.03427

Business Insight: Comparing profit margins across segments helps identify the most efficient and profitable customer groups.

Query 20: Which segment has the lowest profitability despite high sales?

Output:



The screenshot shows a MySQL database interface. On the left, the 'SCHEMAS' panel displays a tree view with 'mysqlproject1' expanded, showing 'Tables' and 'ss_store'. Below this, the 'Table: ss_store' is detailed with its columns and data types. The main window shows a SQL query in the editor, and the 'Result Grid' at the bottom displays the query results.

```
1 select segment,
2     format(sum(profit),2) as 'Total Profit',
3     format(sum(sales),2) as 'Total Sales',
4     format(sum(profit)/sum(sales)*100,5) as 'Profit Margin'
5 from ss_store
6 group by segment
7 order by sum(profit)/sum(sales)*100 desc;
8
```

segment	Total Profit	Total Sales	Profit Margin
Home Office	60,298.68	429,653.15	14.03427
Corporate	91,979.13	706,146.37	13.02551
Consumer	134,119.21	1,161,401.34	11.54805

Table: ss_store

Columns:

- Ship_Mode: text
- Segment: text
- Country: text
- City: text
- State: text
- Postal_Code: int
- Region: text
- Category: text
- Sub_Category: text
- Sales: double
- Quantity: int
- Discount: double
- Profit: double

Business Insight: This segment generates strong sales but lower profitability, indicating opportunities to optimize pricing and discount strategies.

Customer Segment Analysis Summary:

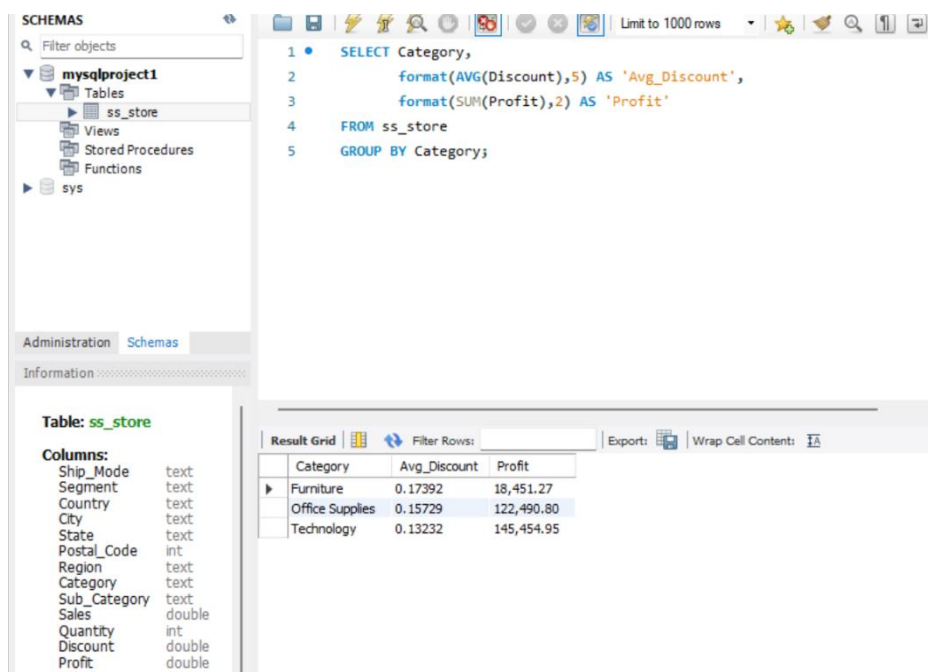
Customer segment analysis revealed the contribution of different customer groups to sales and profitability. The findings help identify high-value customer segments and support targeted business strategies for improving revenue and profit performance.

6.5 Discount Analysis

Discounts play a significant role in driving sales; however, excessive discounting can negatively impact profitability. This section analyses the relationship between discounts and profit performance across products and categories.

Query 21: How does discounting impact profitability across categories?

Output:



The screenshot displays a MySQL database interface. On the left, the 'SCHEMAS' panel shows the 'mysqlproject1' database with tables, views, stored procedures, and functions. The 'ss_store' table is selected. Below this, the 'Table: ss_store' structure is shown with columns: Ship_Mode (text), Segment (text), Country (text), City (text), State (text), Postal_Code (int), Region (text), Category (text), Sub_Category (text), Sales (double), Quantity (int), Discount (double), and Profit (double).

The central query editor shows the following SQL query:

```
1 SELECT Category,
2     format(AVG(Discount),5) AS 'Avg_Discount',
3     format(SUM(Profit),2) AS 'Profit'
4 FROM ss_store
5 GROUP BY Category;
```

The 'Result Grid' on the right shows the output of the query:

Category	Avg_Discount	Profit
Furniture	0.17392	18,451.27
Office Supplies	0.15729	122,490.80
Technology	0.13232	145,454.95

Business Insight: The analysis helps determine whether higher discounts are contributing to increased sales at the expense of profitability.

Query 22: Which products receive the highest average discount?

Output:

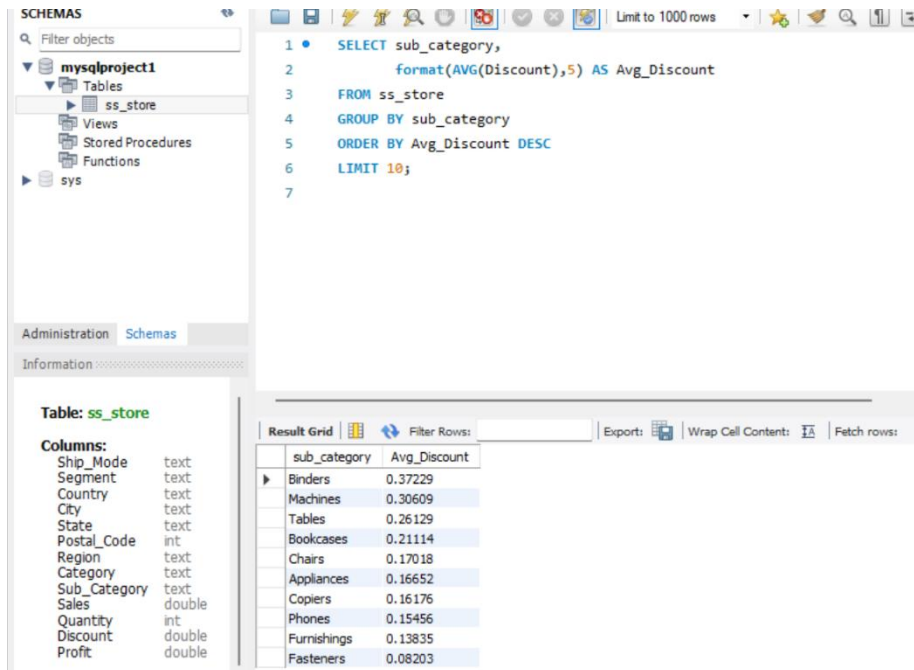


Table: ss_store

Columns:

Ship_Mode	text
Segment	text
Country	text
City	text
State	text
Postal_Code	int
Region	text
Category	text
Sub_Category	text
Sales	double
Quantity	int
Discount	double
Profit	double

Result Grid

sub_category	Avg_Discount
Binders	0.37229
Machines	0.30609
Tables	0.26129
Bookcases	0.21114
Chairs	0.17018
Appliances	0.16652
Copiers	0.16176
Phones	0.15456
Furnishings	0.13835
Fasteners	0.08203

Business Insight: Products receiving higher discounts should be evaluated to ensure they continue to contribute positively to overall business performance.

Query 23: Which categories suffer losses due to excessive discounts?

Output:

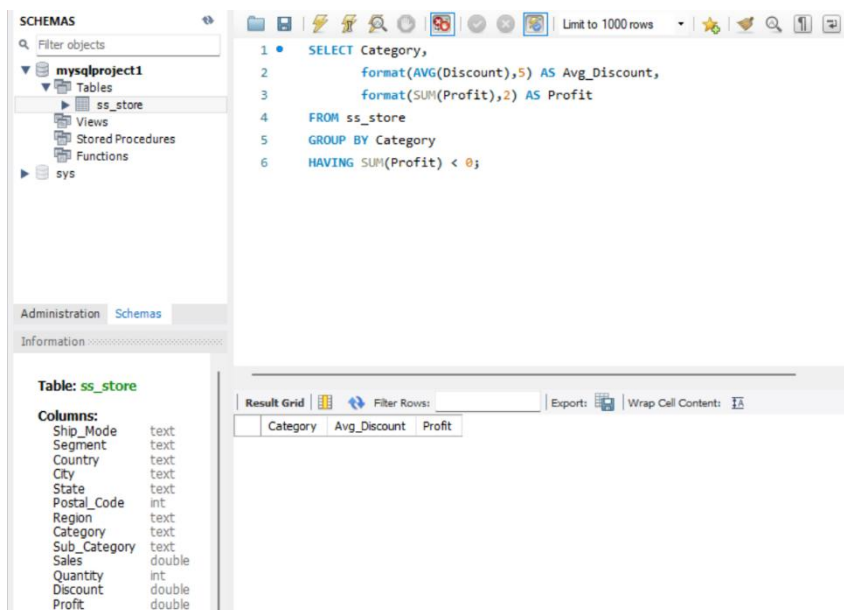


Table: ss_store

Columns:

Ship_Mode	text
Segment	text
Country	text
City	text
State	text
Postal_Code	int
Region	text
Category	text
Sub_Category	text
Sales	double
Quantity	int
Discount	double
Profit	double

Result Grid

Category	Avg_Discount	Profit
----------	--------------	--------

Business Insight: Identifying categories affected by excessive discounting helps management optimize pricing strategies and improve profitability.

Discount Analysis Summary:

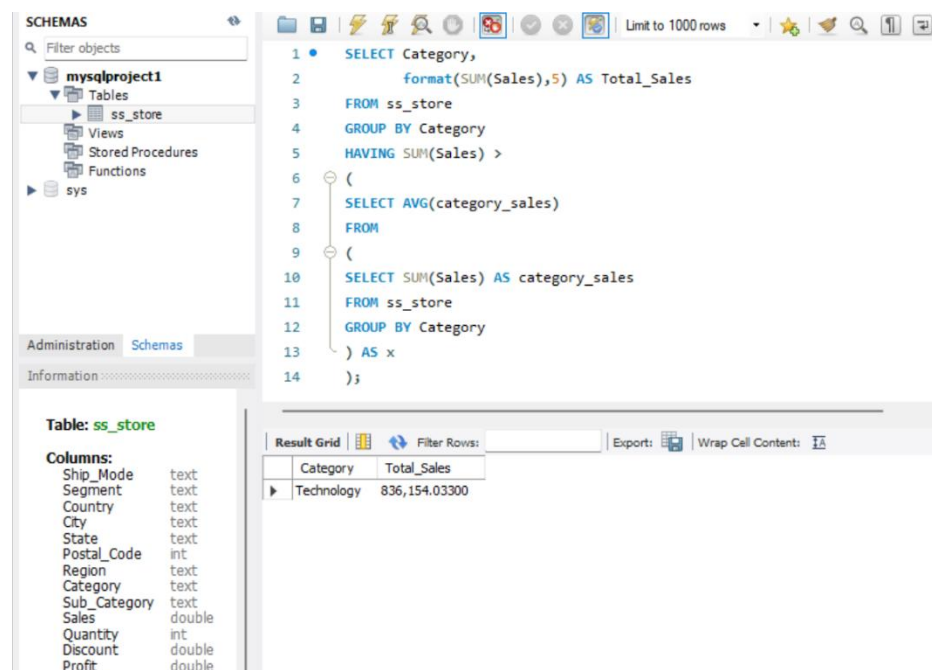
The discount analysis highlighted the relationship between discount strategies and profitability. The findings provide valuable insights for balancing customer incentives with sustainable profit margins.

6.6 Business Growth Insights

Business growth analysis focuses on identifying high-performing categories and regions that contribute significantly to revenue and profitability. These insights help businesses prioritize investments, improve strategic planning, and support future growth initiatives.

Query 24: Identify categories whose sales are above the overall average category sales.

Output:



The screenshot displays a MySQL IDE interface. On the left, the 'SCHEMAS' panel shows a tree view with 'mysqlproject1' expanded, containing 'Tables' (with 'ss_store' selected), 'Views', 'Stored Procedures', 'Functions', and 'sys'. Below this, the 'Table: ss_store' structure is listed with columns: Ship_Mode (text), Segment (text), Country (text), City (text), State (text), Postal_Code (int), Region (text), Category (text), Sub_Category (text), Sales (double), Quantity (int), Discount (double), and Profit (double). The main editor shows a SQL query:

```
1 SELECT Category,
2     format(SUM(Sales),5) AS Total_Sales
3 FROM ss_store
4 GROUP BY Category
5 HAVING SUM(Sales) >
6 (
7     SELECT AVG(category_sales)
8 FROM
9 (
10    SELECT SUM(Sales) AS category_sales
11 FROM ss_store
12 GROUP BY Category
13 ) AS x
14 );
```

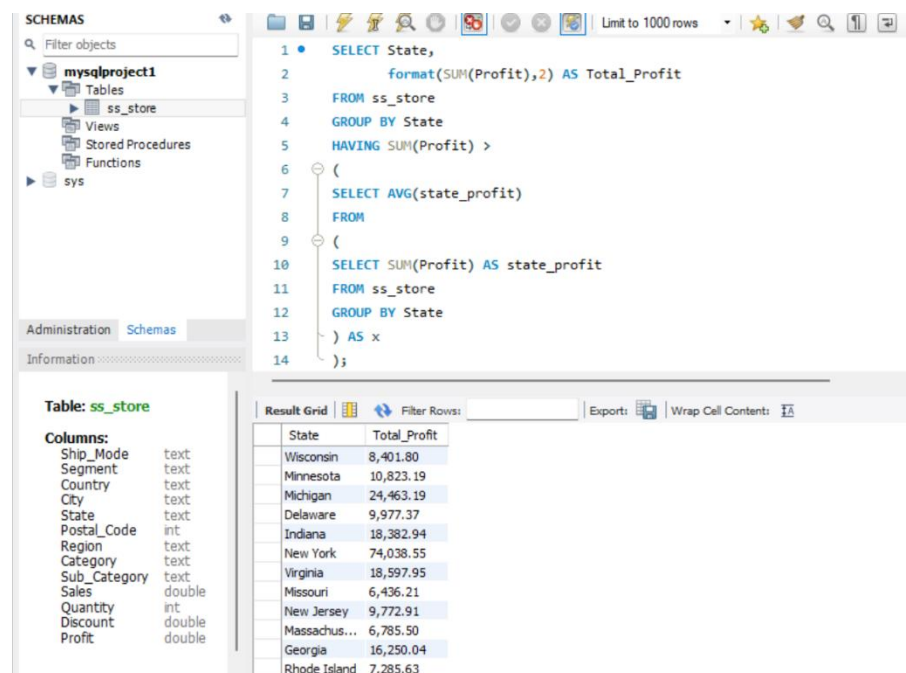
At the bottom, the 'Result Grid' shows the output of the query:

Category	Total_Sales
Technology	836,154.03300

Business Insight: Categories performing above the average sales level represent strong business opportunities and contribute significantly to overall revenue growth.

Query 25: Identify states whose profits are above the overall average state profit.

Output:



The screenshot shows a SQL IDE interface. On the left, a 'SCHEMAS' pane displays a database structure with 'mysqlproject1' containing 'ss_store' and 'sys'. The main editor shows a SQL query that selects states with total profit greater than the average state profit. Below the editor, a 'Table: ss_store' schema is listed. At the bottom, a 'Result Grid' displays the query results as a table with two columns: 'State' and 'Total_Profit'.

```
1 • SELECT State,
2     format(SUM(Profit),2) AS Total_Profit
3 FROM ss_store
4 GROUP BY State
5 HAVING SUM(Profit) >
6 (
7     SELECT AVG(state_profit)
8 FROM
9 (
10    SELECT SUM(Profit) AS state_profit
11 FROM ss_store
12 GROUP BY State
13 ) AS x
14 );
```

State	Total_Profit
Wisconsin	8,401.80
Minnesota	10,823.19
Michigan	24,463.19
Delaware	9,977.37
Indiana	18,382.94
New York	74,038.55
Virginia	18,597.95
Missouri	6,436.21
New Jersey	9,772.91
Massachus...	6,785.50
Georgia	16,250.04
Rhode Island	7,285.63

Business Insight: States generating above-average profit demonstrate strong market performance and can be considered key regions for future business expansion.

Business Growth Insights Summary

The business growth analysis identified high-performing categories and profitable states that contribute significantly to organizational success. These findings provide a foundation for strategic planning and future growth opportunities.

7. Key Findings and Insights

Based on the SQL analysis performed on the Superstore dataset; the following key business insights were identified:

Sales Performance

- The Technology category generated the highest sales revenue of **\$836,154.03**, making it the leading revenue contributor to the business.
- The Consumer segment generated the highest sales revenue of **\$1,161,401.34**, indicating that it is the most valuable customer segment in terms of sales contribution.
- The West Region recorded the highest sales revenue of **\$725,457.82**, demonstrating strong market demand and business presence.

Profitability Analysis

- The Technology category generated the highest profit of **\$145,454.95**, making it the most profitable category.
- The top profit-generating sub-categories were:
 - Copiers – **\$55,617.82**
 - Phones – **\$44,515.73**
 - Accessories – **\$41,936.64**
- The loss-making sub-categories were:
 - Tables – **-\$17,725.48**
 - Bookcases – **-\$3,472.56**
 - Supplies – **-\$1,189.10**

Regional Performance

- The West Region generated the highest profit of **\$108,418.45**, making it the best-performing region overall.
- The Central Region generated the lowest profit of **\$39,706.36**.
- The Central and South regions performed below the average regional profit, indicating opportunities for improvement.

State Performance

- California was the most profitable state with a total profit of **\$76,381.39**.
- New York was the second most profitable state with a total profit of \$74,038.55.
- Major loss-making states identified in the analysis were:
 - Texas – **-\$25,729.36**
 - Ohio – **-\$16,971.38**
 - Pennsylvania – **-\$15,559.96**

Customer Segment Analysis

- The Consumer segment generated the highest profit of **\$134,119.21**.
- The Home Office segment achieved the highest profit margin of **14.03%**, indicating greater profitability efficiency.
- The Corporate segment received the highest average discount of **15.82%**, which may impact profitability.

Discount Analysis

- Binders received the highest average discount of **37.23%**, followed by Machines (**30.61%**) and Tables (**26.13%**).
- The Furniture category had the highest average discount (**17.39%**) but generated comparatively low profit (**\$18,451.27**), suggesting that excessive discounting may be affecting profitability.

Overall Business Insight

The analysis indicates that business performance is primarily driven by the Technology category, Consumer segment, and West Region. While several products and regions contribute significantly to revenue and profit, certain sub-categories and states continue to generate losses. Optimizing discount strategies and improving performance in underperforming regions can help enhance overall profitability and support future business growth.

8. Business Recommendations

Based on the findings from the analysis, the following recommendations are proposed to improve business performance and profitability:

1. Focus on High-Performing Categories

- Increase investment in the Technology category, as it generates both the highest sales and highest profit.
- Promote high-performing products such as Copiers, Phones, and Accessories through targeted marketing campaigns.

2. Improve Loss-Making Product Performance

- Review the pricing, discount strategies, and operational costs associated with Tables, Bookcases, and Supplies.
- Consider reducing excessive discounting or revising product positioning for these sub-categories.

3. Optimize Discount Strategy

- Monitor products receiving very high discounts, particularly Binders, Machines, and Tables.
- Implement controlled discount policies to maintain a balance between sales growth and profitability.

4. Strengthen Regional Performance

- Continue expanding business activities in the West Region, which demonstrates strong revenue and profit performance.
- Develop targeted improvement strategies for the Central and South regions to increase profitability.

5. Expand in High-Profit States

- Increase marketing and sales efforts in California and New York, as they are the most profitable states.

- Use successful business practices from these states as benchmarks for underperforming regions.

6. Address Loss-Making States

- Conduct detailed investigations into the causes of losses in Texas, Ohio, and Pennsylvania.
- Review regional pricing, discount policies, product mix, and customer demand patterns in these states.

7. Strengthen Customer Segment Strategy

- Continue focusing on the Consumer Segment, as it contributes the highest sales and profit.
- Maintain engagement programs for the Home Office Segment, which achieves the highest profit margin.
- Review discount practices for the Corporate Segment to improve profitability.

8. Support Data-Driven Decision Making

- Establish regular monitoring of sales, profit, and discount metrics using SQL-based reporting.
- Continuously analyse business performance to identify emerging opportunities and risks.

Recommendation Summary

By focusing on high-performing categories, optimizing discount strategies, improving underperforming regions, and strengthening customer segment management, the business can enhance profitability, improve operational efficiency, and support sustainable long-term growth.

9. Conclusion

This project analysed the Superstore dataset using SQL to evaluate sales performance, profitability, regional trends, customer segment behaviour, and the impact of discounts on business performance. Through data cleaning, exploratory data analysis, and business-focused SQL queries, valuable insights were extracted to identify high-performing categories, profitable regions, and areas requiring improvement.

The analysis revealed that the Technology category, Consumer segment, and West region were the primary contributors to business success, while certain sub-categories and states generated losses and require strategic attention. Based on these findings, several recommendations were proposed to improve profitability, optimize discount strategies, and support future business growth.

Overall, this project demonstrates the practical application of SQL in transforming raw business data into meaningful insights and highlights the importance of data-driven decision-making in a retail business environment.

Project Outcome:

This project demonstrates practical SQL skills including data validation, exploratory data analysis, business KPI analysis, and insight generation using real-world retail data.